Dungeon Generator – Guide for Game Designers

This guide explains how to use the Dungeon Generator tool inside the Unity Editor to create and configure procedurally generated dungeons. It is written for game designers with no programming knowledge.

1. Opening the Tool

The tool can be found in the menu bar under:

Praxisarbeit → DungeonGenerator

Once opened, a dedicated window appears with two tabs: **Dungeon Settings** and **Room Settings**.

1.1 Overview: How the Generator Works

The tool runs through three steps automatically, one after another, as soon as you click **Generate Dungeon**:

1. **Room Layout:** The dungeon area is recursively subdivided into rectangular regions (similar to a branching tree).
2. **Room Creation:** A room is created within each of these regions.
3. **Corridors and Classification:** The rooms are connected via corridors. The tool then automatically determines which room is used as the Spawn, Boss, Treasure, Puzzle, Shop, or NPC room.

Once these steps are complete, the finished dungeon is built directly in the scene from the assigned prefabs.

2. "Dungeon Settings" Tab

2.1 References

Dungeon Parent – The GameObject in the scene under which the dungeon is created as a child object. This object must exist, otherwise generation is aborted with an error message.

Room Type Library – The ScriptableObject that defines which prefabs and rules apply to each room type (see "Room Settings" tab below). Without an assigned Room Type Library, no rooms can be spawned. The Room Type Library is a ScriptableObject containing a list of RoomData objects, which are used during dungeon generation to provide the properties and contents of each room type.

Note: Both fields are filled in automatically when the tool is opened, provided an object named "Dungeon Parent" exists in the scene and a Room Type Library exists in the project. Both references can be overridden manually at any time.

2.2 Size and Generation

Rule of thumb: Min Room Size should be noticeably smaller than half of Width or Length, otherwise the dungeon cannot be subdivided in a meaningful way. The tool detects this case automatically and shows a corresponding error message.

Field	Description
Dungeon Width / Length	Sets the size of the dungeon in tiles.
Min Room Size	Defines the smallest allowed edge length of a room region. A larger value results in fewer, but bigger, rooms.

Max Split Depth	Sets how many times the dungeon is recursively subdivided. A higher value produces more, but smaller, rooms. Values of 15 or higher are automatically rejected by the tool for performance reasons.

2.3 Corridors

Corridor Width – Sets the width of the corridors in tiles (Small = 1, Medium = 2, Large = 3).

Recommendation: **Medium** is the most practical choice for most dungeon styles. Large produces very wide corridors that can quickly look out of place.

2.4 Seed

Fixed Seed – When enabled, the same seed is used for every generation. This produces reproducible results, which is especially useful for testing and comparisons.

Seed – The numeric value that drives the randomization. The same seed always produces the same dungeon given identical settings.

Note: If Fixed Seed is disabled, the tool automatically uses a random seed based on the current system time for every generation.

2.5 Room Generation

Random Room Sizes – When enabled, rooms get random sizes within their region, creating a more irregular look. When disabled, rooms fill their region with a fixed inner margin, resulting in a more uniform and predictable appearance.

2.6 Buttons

Generate Dungeon – Validates all settings, generates a new dungeon and builds it in the scene. Any existing dungeon is automatically deleted and replaced by the new one.

Delete Dungeon – Removes all previously spawned objects from the Dungeon Parent. Useful for clearing the scene without immediately generating a new dungeon.

2.7 JSON Import and Export

Export – Saves the current room configuration to disk, along with the dungeon layout if a dungeon has already been generated.

Import – Loads a previously exported configuration or saved dungeon layout and rebuilds the dungeon afterwards. Useful for preserving a specific dungeon state.

Note: An export always overwrites the most recently exported JSON file.

3. "Room Settings" Tab

This tab configures which content is spawned for each room type. The following room types are available:

Room Type	Description
Spawn	The player's starting room
Boss	Boss room, always placed as far as possible from Spawn
Treasure	Treasure room
Puzzle	Puzzle room

Shop	Shop room
NPC	NPC room
Standard	Fallback for any room without a specific assignment

Each room type can be expanded and collapsed via a foldout.

3.1 Room Settings – Fields in Detail

RoomData SO – The ScriptableObject holding the data for this room type. If no RoomData is assigned, the room type is skipped during spawning, or reset to the standard room.

RoomData is a ScriptableObject that stores all data for a room type, including the prefabs used, room height, and spawn rules and content. If the RoomData is registered in the Room Type Library, it is used during dungeon generation to create rooms of that type.

Floor / Wall / Ceiling / Door Prefab – The prefabs used to build this room type's floor, walls, ceiling and doors. If no Ceiling prefab is set, the Floor prefab is automatically used for the ceiling as well.

Wall Height – Sets the height of the walls in tiles. Minimum value: 3.

Spawnable Content Rules – A list of placeable prefabs, each with a minimum and maximum spawn count.

Spawn Chance – Probability (0–1) that this room type is assigned at all during classification.

Max Spawn Amount – Sets the maximum number of rooms of this type allowed per dungeon.

Note: Changes to a RoomData object are applied immediately to the scene if a dungeon already exists. Affected rooms are automatically respawned without needing to run Generate Dungeon again.

3.2 Creating a New RoomData

If a new RoomData object is needed:

Right-click in the Project window → Create → DungeonGenerator → RoomData

Then assign the new object to the RoomData SO field of the desired room type.

3.3 Creating a New Room Type Library

If a completely new library is needed:

Right-click in the Project window → Create → DungeonGenerator → Room Type Library

4. Recommended Workflow

1. **Open Dungeon Settings** – Assign Dungeon Parent and Room Type Library.
2. **Configure size** – Set Dungeon Width, Dungeon Length, Min Room Size and Max Split Depth to match the desired dungeon style.
3. **Set Corridor Width to Medium** – the recommended default value.
4. **Enable Fixed Seed** – if reproducible results are needed.
5. **Configure Room Settings** – In the Room Settings tab, assign a RoomData to each required room type and configure the desired prefabs and spawn rules.
6. **Click Generate Dungeon** – Check the result in the scene. If adjustments are needed, change the settings and click Generate Dungeon again; the previous dungeon is automatically replaced. Delete Dungeon is only needed if you want to clear the scene without generating a new dungeon.

7 **Save your state** – Save a desired state via Export.

5. Common Error Messages

Message	Cause	Solution
“Dungeon Parent not assigned”	No GameObject was assigned to the Dungeon Parent field	Assign an appropriate GameObject or create a new one.
“Room Type Library not assigned”	No library was assigned in the Room Type Library field	Select an existing library or create a new one via the Create menu.
“Width and Length must be > 0”	An invalid dungeon size was entered	Enter a number greater than 0 for both values.
“Min Room Size is too large for the dungeon size”	Min Room Size is greater than or equal to half of the smallest dungeon edge	Reduce Min Room Size or increase the dungeon size.
“Max Split Depth is very high”	The Max Split Depth value is 15 or higher	Reduce the value; 10 or lower is recommended.
“Dungeon size is too large”	Width x Length is 100,000 tiles or more	Reduce the dungeon size, otherwise there is a risk of crashing.

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